

Name.....

Roll No.....

**ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.***(An ISO 9001:2015 Certified College)*

Common to AD & CSE

B. TECH DEGREE FIRST INTERNAL TEST-March 2023

Sixth Semester

CST 306 ALGORITHM ANALYSIS AND DESIGN

Time: 2 hours

Max. Marks: 50

PART A

Answer all questions (3 x 5 = 15 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1 ✓	What are the Characteristics of an Algorithm?	3	K1	CO1 & K3
2 ✓	Differentiate between the Space complexity and Time Complexity	3	K2	CO1 & K3
3 ✓	Find <u>lower bound</u> of running time of quadratic function $f(n) = 3n^2 + 2n + 4$.	3	K3	CO1 & K3
4 ✓	What are the different approaches in solving recurrence relations	3	K1	CO2 & K3
5 ✓	What are the properties of the AVL Tree	3	K2	CO3 & K3

PART B

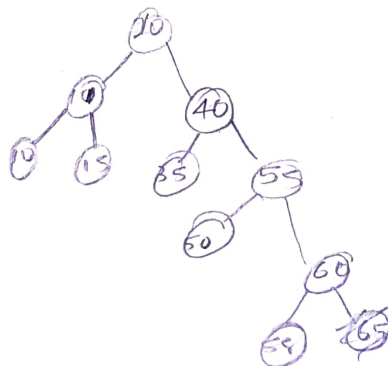
Answer five questions (7 x 5 = 35 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
Answer any two questions				
6	<pre> Linearsearch(arr,n,key) { i = 0; for(i = 0; i < n; i++) { if(arr[i] == key) { printf("Found"); } } } </pre> Find the complexity of the above algorithm using step count method	7	K3	CO1 & K3

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7	Solve the recurrence relation using substitution method $T(n) = \begin{cases} 2T\left(\frac{n}{2}\right) + n & , \text{ otherwise} \\ 1 & , \text{ if } n = 1 \end{cases}$	7	K3	CO2 & K3
8	Solve the recurrence relation using Master's Theorem $T(n) = 2T\left(\frac{n}{2}\right) + \frac{n}{\log^2 n}$	7	K3	CO2 & K3
Answer any two questions				
9	Explain about the various AVL Tree rotations with neat diagrams	7	K1	CO3 & K3
10	Construct AVL tree for the following data 21, 26, 30, 9, 4, 14, 28, 18, 15, 10, 2, 3, 7	7	K3	CO3 & K3
11	What are the different classification of edges that can be encountered during DFS operation and how it is classified? Explain with an example.	7	K3	CO3 & K3
Mandatory Question				
12	Explain in detail about the Master's theorem in solving the recurrence relations	7	K1	CO2 & K3



Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
B.Tech Degree S6 (R, S) / S6 (PT) (R) Examination June 2023 (2019 Scheme)

Course Code: CST306

Course Name: ALGORITHM ANALYSIS AND DESIGN

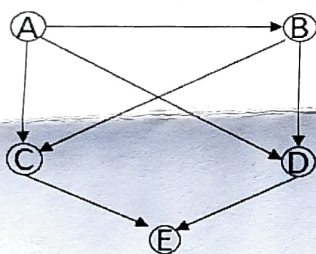
Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
|---|--|-----|
| 1 | Show that for any real constants a and b, where $b > 0$, $(n + a)^b = O(n^b)$ | (3) |
| 2 | Solve the following recurrence equations using Master theorem. | |
| | a. $T(n) = 3T(n/2) + n^2$ | (3) |
| | b. $T(n) = 2T(n/2) + n \log n$ | |
| 3 | Define AVL tree. Explain the rotations performed for insertion in AVL tree. | (3) |
| 4 | Find the different topological ordering of the given graph. | (3) |



- | | | |
|----|---|-----|
| 5 | Write the control abstraction of divide and conquer strategy. | (3) |
| 6 | Compare Strassen's matrix multiplication with ordinary matrix multiplication. | (3) |
| 7 | Differentiate backtracking technique from branch and bound technique. | (3) |
| 8 | What is Principle of Optimality? | (3) |
| 9 | Differentiate P and NP problems. Give one example to each. | (3) |
| 10 | Define graph coloring problem. | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- | | | |
|----|--|-----|
| 11 | a) Define Big Oh, Big Omega and Theta notations and illustrate them graphically. | (7) |
| | b) Find the time complexity of following code segment | (7) |
| | (i) for (int i = 1; i <= n; i *= c) { | |
| | // some O(1) expressions | |
| | } | |

```

for (int i = n; i > 0; i /= c) {
    // some O(1) expressions
}
(ii) for (int i = 1; i <= n; i += c) {
    // some O(1) expressions
}
for (int i = n; i > 0; i -= c) {
    // some O(1) expressions
}

```

OR

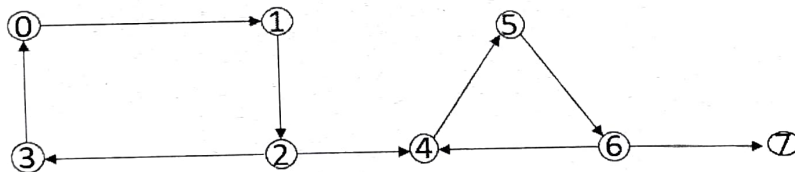
- 12 a) Find the best case, worst case and average case time complexity of binary search. (7)
- b) Find the time complexity of following function using recursion tree method. (7)
- (i) $T(n) = 2T(n/2) + n^2$
- (ii) $T(n) = T(n/3) + T(2n/3) + n$

Module II

- 13 a) Construct AVL tree by inserting following elements appeared in the order. (7)
- 21, 26, 30, 9, 4, 14, 28, 18, 15
- b) Explain union and find algorithms in disjoint datasets. (7)

OR

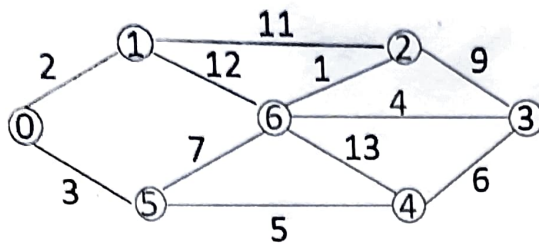
- 14 a) Write DFS algorithm for graph traversal. Also derive its time complexity. (7)
- b) Find the strongly connected components of the given directed graph. (7)

**Module III**

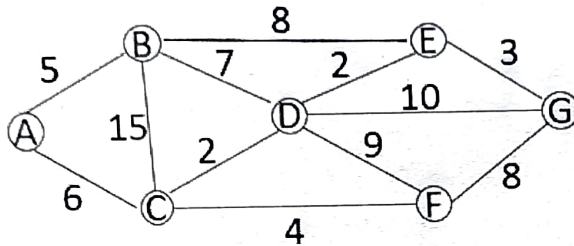
- 15 a) Explain 2-way merge sort algorithm with an example and derive its time complexity. (7)
- b) Find the optimal solution for the following Fractional Knapsack problem. (7)
- Given the number of items $(n) = 7$, capacity of sack $(m) = 15$,
 $W = \{1, 3, 5, 4, 1, 3, 2\}$ and $P = \{10, 15, 7, 8, 9, 4\}$

OR

- 16 a) Apply Kruskal's algorithm for finding minimum cost spanning tree. (7)



- b) Apply Dijkstra's algorithm for finding the shortest path from vertex A to all other vertices. (7)

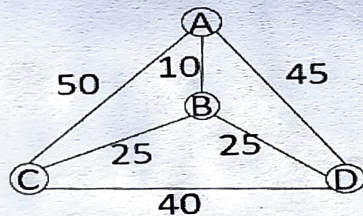


Module IV

- 17 a) Find the optimal parenthesis of matrix chain product whose sequence of dimensions is $5 \times 4, 4 \times 6, 6 \times 2, 2 \times 7$ (8)
 b) Explain 4 queen problem. Draw the state space tree for 4 queen problem. (6)

OR

- 18 a) Define TSP problem. Apply branch and bound algorithm for solving TSP. (9)



- b) Write Floyd Warshall's algorithm for finding all pairs shortest path algorithm. (5)

Module V

- 19 a) Explain the first fit-decreasing strategy of bin packing algorithm. (7)
 b) Prove that Clique Decision problem is NP-complete. (7)

OR

- 20 a) Differentiate Las Vegas and Monte Carlo algorithms (7)
 b) Explain randomized quick sort with the help of suitable examples. (7)
